

Project Design Phase

Proposed Solution Template

Date	29 June 2026
Project Name	OptiCrop
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers often struggle to select the most suitable crop due to changing weather conditions, varying soil quality, and limited access to data-driven agricultural guidance. This results in lower crop yields, inefficient use of fertilizers and water, increased production costs, and reduced profitability.
2.	Idea / Solution Description	OptiCrop – Smart Agricultural Production Optimization Engine is an intelligent web/mobile application that uses Machine Learning to analyze soil nutrients (Nitrogen, Phosphorus, Potassium), soil pH, temperature, humidity, rainfall, and other environmental factors to recommend the most suitable crop. The system provides accurate, data-

		driven recommendations that help farmers improve productivity, optimize resource utilization, and make informed farming decisions.
3.	Novelty / Uniqueness	The solution combines multiple agricultural parameters with Machine Learning to generate personalized crop recommendations. Unlike traditional methods that rely on general advice or experience, OptiCrop provides real-time, data-driven recommendations tailored to each farm's environmental conditions, making crop planning more accurate and efficient.
4.	Social Impact / Customer Satisfaction	The project empowers farmers with reliable agricultural insights, leading to increased crop productivity, reduced resource wastage, and improved income. It promotes sustainable farming practices while enhancing customer satisfaction through easy-to-use technology, accurate recommendations, and better decision-making support.
5.	Business Model (Revenue Model)	The platform can operate on a freemium model , where basic crop recommendations are available free of cost, while advanced analytics, premium advisory services, weather forecasting,

		fertilizer optimization, and farm reports are offered through subscription plans. Additional revenue can be generated through partnerships with agricultural organizations, fertilizer companies, seed suppliers, and government agricultural programs.
6.	Scalability of the Solution	OptiCrop is designed to be highly scalable. It can be expanded to support multiple crops, different soil types, various geographical regions, multiple languages, IoT-based sensor integration, satellite data, weather APIs, and AI-powered disease prediction. The cloud-based architecture enables the platform to serve thousands of farmers simultaneously with minimal performance impact.